

read and poll FC sensors, analogue sensors, and specialty sensors like LEGO's ultrasonic sensors.

The BrickPi has a power unit that allows you to power the BrickPi and Raspberry Pi with a battery (rather than the usual wall plug), so you can make mobile robots. The Raspberry Pi, with two USB ports, supports WiFi and Bluetooth to make your robot wireless.

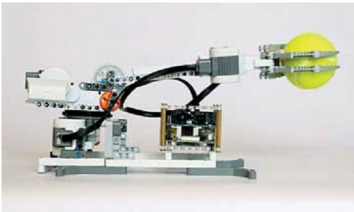
The BrickPi connects five sensors and four motors using LEGO's easy-to-attach sensor cabling system. It also includes an enclosure that LEGO technical parts, sensors and motors can be attached to.

"We have put a lot of work into the software package for the Raspberry Pi as well. This includes easy-to-follow examples for robotic cars, turrets, tanks and arms. We have drivers for 'C,' Python and MIT's graphic language Scratch. The beauty of the hardware architecture and the Raspberry Pi is that any language that runs on the Raspberry Pi can be used to control a student's robot," says Cole.

All of these features, working together, allow students to quickly and easily put together a robot that can be Web-controlled, USB-enabled and language-neutral.

Challenges faced

To come up with this kind of a platform, the team must have faced a host of challenges. Cole shares, "We got a



BrickPi as a robot arm

lot of great feedback from our backers on the KS site. They requested a few basic changes. Mid-campaign, we redesigned the entire BrickPi, basically doubling the functionality, improving the power system and opening it up to a lot of other sensors and projects, all while holding the price down. We were really proud of ourselves for redeveloping the whole thing in a two-week timespan."

Cole adds, "It's also an entirely open source project, which has the challenge or advantage of having everyone look at it, criticise it and question the choices. If handled properly, well and good. But if handled poorly, it can be a disaster for the product."

From Raspberry Pi to BrickPi

When enquired about the time taken in getting this idea from inception to fruition, Cole informs, "We started working on it in February 2013. We put it down for a month, then picked it up. So we've probably been working on it for about four months total."

Why Kickstarter?

By choosing the crowd-funding route, the developers primarily wanted to build a community around this product. They believe Kickstarter is a great way to find interested people, build community and get feedback. They also believe that the community will help bring the price of the product down to the cost of the Raspberry Pi.

But since they've come a lot further, they have added a series of stretch funding goals to explain what they plan to do with the extra money raised. Some of them are listed in the box below.

"We've put the BrickPi up for pre-sale on our website. Once our promises to the Kickstarter backers are fulfilled and all units shipped, we'll start shipping the presales and regular sales of the BrickPi," says Cole.

When asked whether there were any other products of this kind available in the market, Cole informs, "Not yet, but some products do some similar things. There are a number of companies trying to put together erector sets/robotic sets that use the Raspberry Pi as the brain."

In the future

Talking about their future plans, Cole shares, "We think the hardware is finished, but no product survives first contact with the customer. So we're looking forward to getting a few thousand pieces of feedback from the initial batch, and using it to improve the design in another iteration or two. The software will probably never be done, but that's what makes our section of the market really interesting."

"It also makes this product so fun. We are going to get a lot of feedback on how it is used, and we're going to make improvements accordingly," Cole adds. ●

The author is a tech correspondent at EFM Bengaluru

Stretch goals

\$100,000—4 ports in a Storm. The team will add the fifth sensor port and the fourth motor port to everyone's BrickPi. Right now the board has a layout for them. When the funding reaches \$100K, every board will be shipped with all the sensor ports filled.

\$125,000—Seafarer. The team will develop a custom project to put the BrickPi into a solar-powered boat and attempt to sail it around the world. Like the Space Race, it will produce a highly entertaining video to help publish full plans for building the boat. It will equip the boat with satellite messaging and set up a website where one can track the boat in real time.

\$150,000—Destroy the 9V. All BrickPi-level backers will get a 6xAA-size battery pack. There will be no more 9V power supplies.